

Analogue Front End

Content

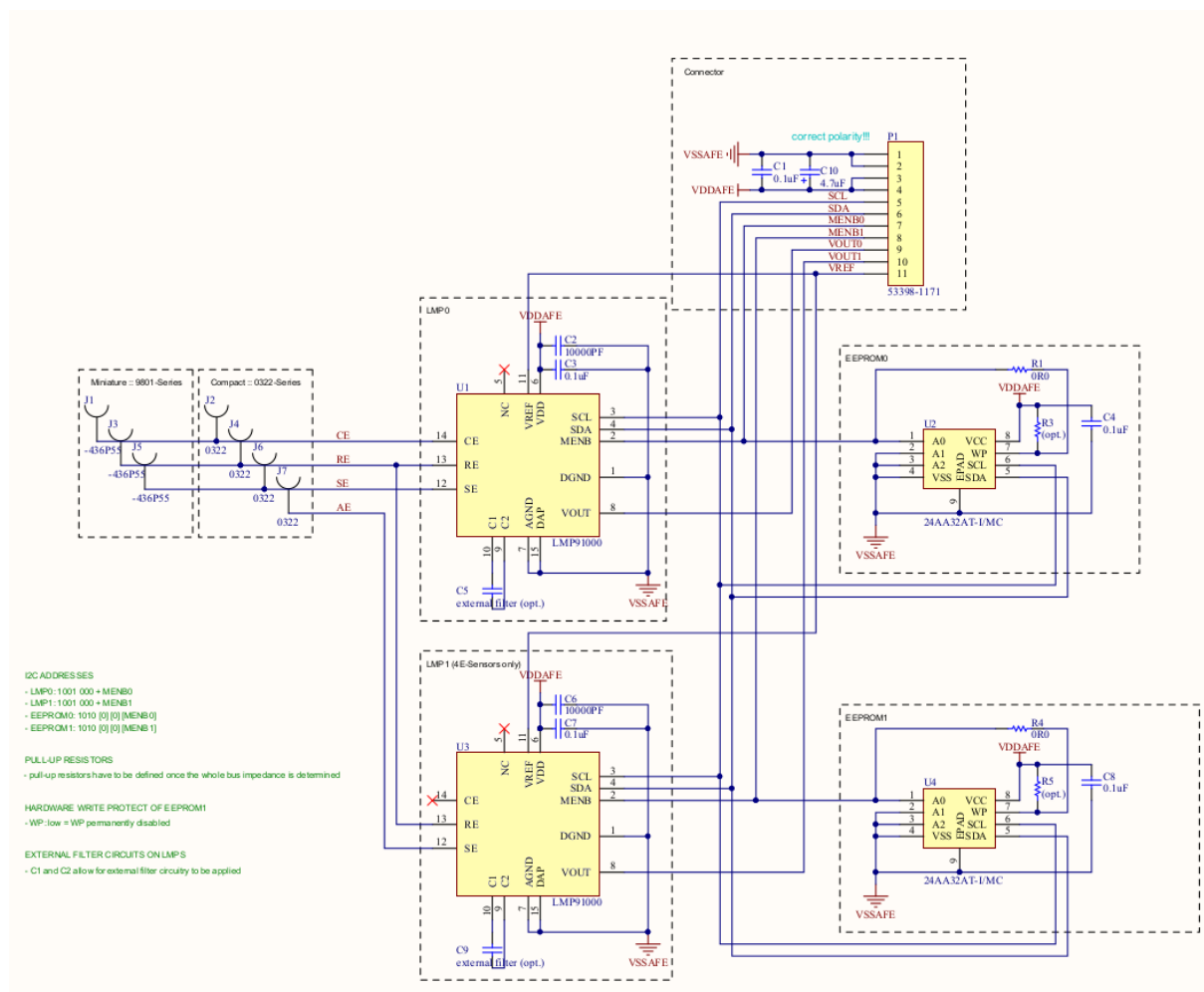
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1. Overview

XXX

1.1. Blockdiagram

ToDo: simple block diagram



1.2. Connector Pin-out

Pin	Name	Description
1	VDDAFE	Supply voltage for AFE
2	SCL	Clock for I2C bus
3	SDA	Data for I2C bus
4	MENB0	Selection Pin for LMP's and EEPROMs
5	MENB1	Selection Pin for LMP's and EEPROMs
6	VDDMCU	Supply voltage for
7	VSSAFE	Ground for AFE
8	VOUT0	Analog output voltage of LMP0
9	VOUT1	Analog output voltage of LMP1
10	VREF	Reference voltage for AFE (typ. 2.5V)
11	VSSAFE	Ground for AFE

2. I2C-Interface

Both EEPROMs as well as the AFE Potentionstats are connected to the same I2C-bus. To select which LMP to communicate with, the signals MENB0 and MENB1 are used:

Table 1: LMP selection

MENB1	MENB0	Active LMP
0	0	Invalid configuration
0	1	LMP1 is selected
1	0	LMP0 is selected
1	1	Neither LMP is selected

The same signals are also used to select the EEPROM. Additionally, MENB0 and MENB1 are used to protect the EEPROM from unintentional write access:

MENB0	MENB1	EEPROM0	EEPROM1
0	0	Invalid configuration	
0	1	I2C-address 0x50 (writable)	I2C-address 0x51 (write protected)
1	0	I2C-address 0x51 (write protected)	I2C-address 0x50 (writable)
1	1	Invalid configuration (both EEPROM's are write protected)	

2.1. LMP communication

In order to read or write from an LMP register, the following procedure has to be followed:

1. Set MENBx low (see Table 1: LMP selection)
2. Generate a start-condition on the I2C-bus, followed by the 7 Bit slave-address of the LMP (0x90) and the read/write-bit set to low.
3. Transmit the internal register address of the register to be read/written (see chapter 3).
4. Depending on whether it's a read or write access:
 - a. Write-access: send the register-value to be written
 - b. Read-access: generate a Repeat-Start-Condition on the bus, followed by the 7 bit slave-address with the read/write-bit set to high, and read the register-value from the LMP.

3. LMP Registers

Both LMP91000 Potentiostats provide the following registers:

Address	Name	Power on default	Access	Lockable?
0x00	STATUS	0x00	Read-only	No
0x01	LOCK	0x01	R/W	No
0x02 ... 0x09	Reserved	n/a	n/a	n/a
0x10	TIACN	0x03	R/W	Yes
0x11	REFCN	0x20	R/W	Yes
0x12	MODECN	0x00	R/W	No
0x13 ... 0xFF	reserved	n/a	n/a	n/a

3.1. STATUS – Address 0x00

The status bit is an indication of the LMP91000's power-on status. If its readback is "0", the LMP91000 is not ready to accept other I2C commands.

Bit	Name	Function
[7:1]	RESERVED	
0	STATUS	Status of Device 0 Not Ready (default) 1 Ready

3.2. LOCK – Address 0x01

The lock bit enables and disables the writing of the TIACN and the REFCN registers. In order to change the content of the TIACN and the REFCN registers the lock bit needs to be set to "0".

Bit	Name	Function
[7:1]	RESERVED	
0	LOCK	Write protection 0 Registers 0x10, 0x11 in write mode 1 Registers 0x10, 0x11 in read only mode (default)

3.3. TIACN – Address 0x10

The parameters in the TIA control register allow the configuration of the transimpedance gain (RTIA) and the load resistance (RLoad).

Bit	Name	Function
[7:5]	RESERVED	
[4:2]	TIA_GAIN	TIA feedback resistance selection 000 External resistance (default) 001 2.75kΩ 010 3.5kΩ 011 7kΩ 100 14kΩ 101 35kΩ 110 120kΩ 111 350kΩ
[1:0]	RLOAD	RLoad selection 00 10Ω 01 33Ω 10 50Ω 11 100Ω (default)

3.4. REFCN – Address 0x11

The parameters in the Reference control register allow the configuration of the Internal zero, Bias and Reference source. When the Reference source is external, the reference is provided by a reference voltage connected to the VREF pin. In this condition the Internal Zero and the Bias voltage are defined as a percentage of VREF voltage instead of the supply voltage.

Bit	Name	Function
7	REF_SOURCE	Reference voltage source selection

		0 Internal (default) 1 external
[6:5]	INT_Z	Internal zero selection (Percentage of the source reference) 00 20% 01 50% (default) 10 67% 11 Internal zero circuitry bypassed (only in O2 ground referred measurement)
4	BIAS_SIGN	Selection of the Bias polarity 0 Negative (VWE – VRE)<0V (default) 1 Positive (VWE –VRE)>0V
[3:0]	BIAS	BIAS selection (Percentage of the source reference) 0000 0% (default) 0001 1% 0010 2% 0011 4% 0100 6% 0101 8% 0110 10% 0111 12% 1000 14% 1001 16% 1010 18% 1011 20% 1100 22% 1101 24%

3.5. MODECN – Address 0x12

The Parameters in the Mode register allow the configuration of the Operation Mode of the LMP91000.

Bit	Name	Function
7	FET_SHORT	Shorting FET feature 0 Disabled (default) 1 Enabled
[6:3]	RESERVED	
[2:0]	OP_MODE	Mode of Operation selection 000 Deep Sleep (default) 001 2-lead ground referred galvanic cell 010 Standby 011 3-lead amperometric cell 110 Temperature measurement (TIA OFF) 111 Temperature measurement (TIA ON)

4. EEPROM

Each AFE contains two EEPROMS with 32kbits (4kBytes) each. They can be used to store calibration or configuration data for the host MCU. The AFE itself, i.e. the Potentiostats do not make use of the EEPROM's themselves.

Please note that certain address ranges of the EEPROM's contain manufacturer programmed calibration data.

Reserved Range for EEPROM0 and EEPROM1:

From	To	Content
0x00	0x200	Factory calibration data

4.1. Memory map of EEPROM 0

Address	Content
0x00	TIACN of LMP0
0x01	TIACN of LMP1
0x02	REFCN of LMP0
0x03	REFCN of LMP1
0x04	MODECN of LMP0
0x05	MODECN of LMP1
0x06	Number of pins (3 or 4 electrodes)
0x07	Selection of ppm calculation
0x08	Current loop unit
0x09	Ppm @ 20mA
0x0d	Loop-current at 20mA
0x11	Enable buzzer-alarm
0x12	Alarm trigger-value
0x16	External reference voltage
0x80	Calibration: IOut LMP0 @ GA0 and GB0
0x84	Calibration: IOut LMP1 @ GA0 and GB0
0x88	Calibration: IOut LMP0 @ GA1 and GB1
0x8C	Calibration: IOut LMP1 @ GA1 and GB1
0x90	Calibration: IOut LMP0 @ GA2 and GB2
0x94	Calibration: IOut LMP1 @ GA2 and GB2